

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a practical framework for improving the reasoning behavior of large language models through rule-based outcome supervision. However, widely used RLVR optimizers such as GRPO can underuse generated data: fixed token-level ratio clipping suppresses informative updates, while step-local advantage standardization assigns zero gradients to response groups with identical rewards. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization procedure designed to address these response- and token-level inefficiencies within standard RLVR training. DCPO combines a token-specific dynamic clipping rule, which adapts the admissible probability-ratio interval according to old token probability and preserves more update room for low-probability tokens, with smooth advantage standardization, which incorporates cumulative reward statistics to retain response-level learning signals across training steps. It further uses an only-token-mean loss that avoids length-induced weighting artifacts across responses. Across four mathematical reasoning benchmarks and four Qwen2.5-Math/Qwen2.5 backbones, DCPO obtains the strongest overall performance among the evaluated methods. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 under greedy decoding and 38.8 Avg@32 under 32-sample decoding on AIME24, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B, DCPO achieves 23.3/19.0 on AIME25, outperforming GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). DCPO also improves the nonzero-advantage ratio by an average of 28% over GRPO across four models, doubles training efficiency relative to DAPO under the reported update schedule, and lowers the token clipping ratio by an order of magnitude compared with GRPO and DAPO while maintaining higher benchmark performance. These results show that, in math-only RLVR with rule-based rewards, DCPO converts generated samples into useful optimization signal more effectively than the evaluated baselines.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central target of recent post-training research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) offers a direct route to this objective because it optimizes models using rule-based outcome rewards rather than preference labels alone. This formulation is especially appropriate when correctness can be checked automatically and when successful behavior requires multi-step deliberation, as in mathematical problem solving and code generation (DeepSeek-AI et al., 2024).

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) has become an influential foundation for RLVR training. Its empirical value is clear, yet subsequent analyses, including DAPO (Yu et al., 2025), identify limitations that are consequential precisely in reasoning-oriented optimization. GRPO may enter entropy collapse, with advantage values decaying toward zero in later training, and its fixed clipping rule can remove rare but informative token-level learning signals, thereby narrowing the policy’s effective exploration range.

DAPO addresses these issues through Clip-Higher, which relaxes the upper clipping bound to mitigate entropy collapse, and Dynamic Sampling, which filters uniformly rewarded samples so that each batch contains nonzero gradients. These interventions improve the available training signal, but they

do not eliminate two structural inefficiencies. First, **response-level inefficiency** remains because dynamic sampling expends additional generation budget to replace filtered responses. Second, a **token-level clipping deficit** remains because the admissible ratio interval should depend on the old probability of each token; a uniform upper bound can therefore be unnecessarily restrictive in informative low-probability regions. Related methods that avoid directly discarding zero-gradient responses still leave this token-level mismatch unresolved.

We propose **Dynamic Clipping Policy Optimization (DCPO)** as a unified policy-optimization pipeline for these response- and token-level failures. The first component is **smooth advantage standardization across cumulative steps**: by combining the current reward distribution with the cumulative distribution of previously generated responses, DCPO prevents identically rewarded responses from becoming unusable zero-advantage samples (see Section 2). This mechanism preserves response-level optimization signal without altering the verifiable-reward interface. The second component is **dynamic clipping based on old token-specific probabilities**. Instead of assigning a single fixed interval to all tokens, DCPO widens the admissible ratio range for lower-prior-probability tokens and thereby allocates additional update capacity to token-level exploration (see Section 2 and Appendix A.6). The third component is an only-token-mean loss: DCPO averages over tokens within each response rather than across the full batch, preserving the relative advantage structure among responses to the same prompt and avoiding the dilution introduced by batch-level averaging.

We evaluate DCPO on a merged mathematical training corpus formed from Math-DAPO-17k (Yu et al., 2025) and levels 3-5 of the MATH dataset (Liu et al., 2025), and we test the resulting policies on four standard mathematical reasoning benchmarks. The experiments span four model backbones and compare against GRPO, DAPO, and, where available, GSPO under the same reported training schedule. DCPO is the strongest evaluated optimizer in this setting, with especially clear gains on the high-difficulty AIME benchmarks: on AIME24-Avg@32 with 32 sampled responses, DCPO-7B scores 38.8 compared with 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO); on AIME25-Avg@32, DCPO-14B reaches 19.0 compared with 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). DCPO also raises the average nonzero-advantage ratio by 28% over GRPO, halves the training cost relative to DAPO for the same update count, and reduces token clipping by an order of magnitude while improving task performance.

Appendix A.2 collects the notation, advantage definitions, and baseline objectives for GRPO, DAPO, and GSPO.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a standard mechanism for stabilizing reinforcement-learning updates because it limits parameter changes induced by large importance weights. This consideration is particularly important in off-policy estimation. When an estimate is reweighted by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, the estimator for $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$ remains unbiased, but its variance can increase substantially, as shown in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To control this variance inflation, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio by a fixed clipping hyperparameter ϵ , as in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping interval, however, constrains exploration most strongly in regions where the old policy assigns low probability mass. In these regions, the absolute room for probability reduction or expansion is not matched to the potential informativeness of a token, so meaningful low-probability updates may be clipped. Asymmetric clipping in methods such as DAPO partially relaxes this constraint, but it still applies a uniform bound that does not adapt to token-specific probabilities. GSPO moves the constraint to the sequence level; when the sequence-level importance ratio exceeds asymmetric bounds, discarding an entire response can remove informative tokens together with high-variance ones.

From a variance–bias perspective, this motivates a constraint that depends on probability mass rather than only on a globally fixed ratio interval. We therefore use the probability-dependent restriction in Equation (3) as the basis for the clipping rule.

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero gives a closed-form admissible interval for $r(x)$. The resulting **dynamic-adaptive clipping bounds** vary with the old probability $q(x)$ and with the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This construction expands the policy’s admissible update range in low-probability regions while retaining tighter control where the old policy is already confident. Following the motivation of dual clipping (Ye et al., 2020), we cap the maximum bound for both positive and negative advantages at 10 to avoid excessively large ratio values and overtraining. Appendix A.5 and Appendix A.6 provide the derivation and the accompanying importance-sampling analysis.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ for the j -th response is obtained by standardizing the rewards $R_{j=1,\dots,G}^i$ generated for the same prompt at step i , as shown in Equation (11) in Appendix A.3. This step-local normalization creates two training pathologies. First, when stochastic response generation produces identical rewards within a step, all corresponding advantages become zero, so the prompt contributes no parameter update even if the sampled reasoning trajectories differ. Second, high-entropy sampling can produce skewed label counts, inducing large cross-step fluctuations in standardized advantages and, in some cases, sign reversals that destabilize optimization.

Recent evidence (Yue et al., 2025) further suggests that RLVR primarily refines a policy by adjusting the likelihood of already reachable correct samples rather than by fundamentally replacing the underlying response distribution. Under this view, responses to the same prompt can be treated as samples from a shared reward distribution accumulated across training, motivating the *cumulative* standardization in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the normalization statistics are computed over all responses generated so far for the same prompt.

To reduce instability in both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we define two smoothed quantities, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. Each quantity is a step-dependent weighted average of the two standardized advantages, with weights determined by the training step i as in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training proceeds, allowing the advantage estimate to follow the latest policy outputs. Conversely, $\hat{S}A_{total,j}^i$ increasingly emphasizes the cumulative distribution, interpreting the current-step deviation as a local adjustment within the broader reward history. The final advantage $\hat{A}_j^i$ is the smoothed estimate with the smaller absolute

value, which limits the influence of fluctuations introduced by either normalization scheme and stabilizes the update as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt enters optimization, later responses for the same prompt can still participate in parameter updates even when the step-local advantage would be zero. Thus, when identical rewards occur within a step, the corresponding responses receive update signal through the cumulative term $\frac{1}{i}\hat{A}_{total,j}^i$, preserving useful response-level information that would otherwise be discarded.

3.1 OTM LOSS

Although the **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) balances token contributions across the batch, it can distort the intended relationship among responses of different lengths. Consider two responses to the same prompt: response A has advantage 1 and length 500, whereas response B has advantage 0.5 and length 1500. Under TLM, the loss for A is weighted by $\frac{500}{500+1500} = 0.25$, and the loss for B by $\frac{1500}{500+1500} = 0.75$. The resulting total contributions are therefore $1 \times 0.25 = 0.25$ for A and $0.5 \times 0.75 = 0.375$ for B , so the lower-advantage response exerts greater influence solely because it is longer.

Conversely, the **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO) averages across the G responses and thereby attenuates the standardized advantage scale. This effect becomes more pronounced as the response group grows. Since standardized advantages already encode relative relationships among responses, DCPO averages only over the tokens within each response. Dividing by $|o_i|$ distributes a response’s importance uniformly across its own tokens while preserving the relative advantage structure across responses. We refer to this aggregation as the **Only Token Mean loss (OTM)**, defined in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, the DCPO loss does not include an explicit KL-divergence penalty. The lower and upper clipping thresholds in Equation (8) are instantiated with the dynamic clipping bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate four backbones: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training uses a merged corpus of approximately 25k mathematical problems, combining DAPO-Math-17K (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). All training and evaluation prompts use the Qwen-Math template described in Appendix A.8.

Evaluation We evaluate on four standard mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, we report Avg@1 because the benchmark contains 500 problems. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets, we report both Avg@1 and Avg@32 to characterize greedy accuracy and sampled-policy behavior.

- **Avg@1:** Greedy-decoding accuracy, measuring the model’s single deterministic prediction for each problem.
- **Avg@32:** Average accuracy over 32 sampled responses per problem, using temperature 1.0 and top-p 1.0, measuring the quality and stability of the sampled policy distribution.

Baseline Settings We compare against GRPO and DAPO on all four model backbones, and include GSPO on Qwen2.5-Math-7B and Qwen2.5-14B as additional baselines. Appendix A.9 reports the clipping thresholds ϵ_* , codebase, and shared training hyperparameters.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Performance across benchmarks. Avg@1: greedy decoding; Avg@32: sampling 32 times. Boldface shows the best values, and Δ shows improvement over baselines.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Table 1 reports benchmark performance for GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25. Avg@1 measures greedy-decoding accuracy, whereas Avg@32 measures average accuracy over 32 sampled responses for AMC23, AIME24, and AIME25. Across the four evaluated backbones, DCPO delivers the best average Avg@1 in all cases and the best average Avg@32 in three of four cases, with the remaining 3B Avg@32 result within 0.4 points of DAPO.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO achieves the highest average scores (47.2/32.8), exceeding GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), improving over GRPO (+1.3/+1.7) and remaining competitive with DAPO, with a higher Avg@1 (+1.0) and a small Avg@32 gap (-0.4). On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On Qwen2.5-14B, DCPO obtains (53.2/39.0), improving over GRPO (+6.5/+7.7), DAPO (+1.3/+0.1), and GSPO (+4.2/+5.5). These comparisons show that DCPO’s gains are not confined to a single backbone or decoding mode, while also preserving the cases in which individual baselines remain strong on specific benchmarks.

Robustness on AIME under Sampling (Avg@32) The AIME24 and AIME25 datasets provide a stringent test of sampled mathematical reasoning. Under Avg@32, DCPO is particularly strong at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B scores 38.8, compared with 32.1 for GRPO, 31.6 for DAPO, and 34.9 for GSPO; on AIME25-Avg@32, DCPO-14B reaches 19.0, compared with 10.5 for GRPO, 15.3 for DAPO, and 9.9 for GSPO. These improvements indicate that DCPO improves the sampled policy distribution on difficult competition-style math benchmarks, where maintaining useful exploration while increasing the probability of correct trajectories is essential.

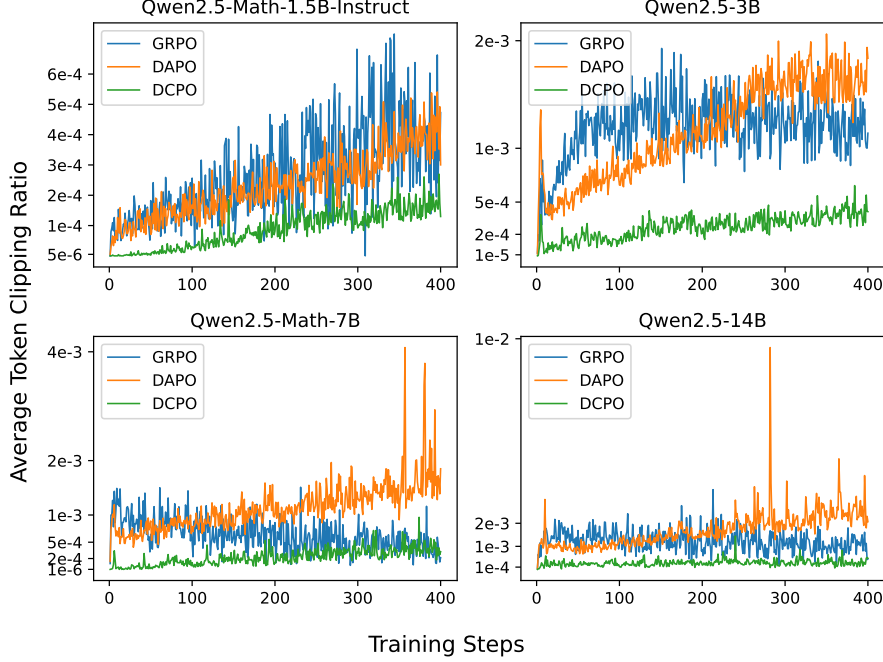


Figure 1: *TCR* across models and methods.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of generated tokens excluded from back-propagated policy updates by the clipping operation. This diagnostic measures how much token-level optimization signal is removed by a policy objective, complementing benchmark accuracy with a direct view of update utilization.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

As shown in Figure 1, GRPO exhibits model-dependent *TCR* trajectories: the ratio increases for Qwen2.5-Math-1.5B and Qwen2.5-3B, but gradually decreases for Qwen2.5-Math-7B and Qwen2.5-14B. DAPO, by contrast, shows an increasing *TCR* across all four models, indicating that progressively more tokens are excluded from updates as training proceeds. Both GRPO and DAPO also show visible fluctuations and occasional spikes. GSPO, evaluated on the 7B and 14B models, has an average *TCR* above 11% and above 15%, respectively, substantially higher than the token-level clipping methods in these runs.

DCPO maintains a substantially lower and more stable *TCR*. Across model sizes and training stages, its token clipping ratio remains approximately an order of magnitude below GRPO and DAPO, meaning that more generated tokens remain eligible to contribute to policy updates. This pattern is consistent with the intended role of dynamic-adaptive clipping: preserve tight control for high-probability tokens while reducing unnecessary clipping of low-probability tokens that may carry useful exploratory signal.

For Qwen2.5-Math-7B, the generated-token distribution rapidly concentrates on high-confidence tokens: after approximately 60 steps, about 95% of generated tokens have probability ($q(x) > 0.9$), and this fraction rises to about 97% after approximately 100 steps. As illustrated in Figure 4d, once the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ participates in model updates within $[\frac{1}{1+\epsilon}, 1]$ without being discarded under the compared clipping rules. Most tokens therefore lie in a region that is rarely clipped, while the later-stage growth of TCR in Figure 1 reflects the increasing importance of the low-probability tail.

Previous work (Wang et al., 2025) identifies **high-entropy tokens**, which are often low-probability tokens, as important drivers of emergent reasoning behavior. Since lower token probability is associated with higher information content in this setting, a wider admissible update range for such tokens increases the effective exploration space available to the policy. The empirical TCR patterns therefore support the design of **dynamic-adaptive clipping**, which relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens.

5.3 THE RESPONSE UTILIZATION RATIO (RUR)

During RLVR training, a generated response contributes to a policy update only when its advantage is nonzero. We quantify this response-level efficiency with the **Response Utilization Ratio (RUR)**, defined as the percentage of generated responses with nonzero advantage.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that GRPO has a low average RUR , with a mean of 43.8% across the four evaluated models. Thus, more than half of generated responses receive zero advantage and do not contribute to the update. DCPO reaches an average RUR of 71.8%, corresponding to an absolute increase of approximately 28 percentage points and a relative improvement of 64% over GRPO.

Table 2: Average RUR over 400 training steps.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 further shows that GRPO’s RUR declines sharply over training across model sizes, falling from above 90% to below 50%; on Qwen2.5-Math-7B, it reaches approximately 30% after about 200 steps. This decline is consistent with the limitation of step-local advantage standardization, under which increasingly many response groups receive identical rewards and therefore zero gradients. DCPO, using smooth advantage standardization, stabilizes near 70% after the first epoch and remains high through subsequent training.

The early RUR above 90% is primarily explained by formatting failures from under-trained models: malformed outputs receive negative rewards, which artificially increases the number of nonzero advantages. From the second epoch onward, RUR remains near 70% rather than reaching 100%. Two cases account for much of the remaining zero-advantage mass. (i) Some math problems are sufficiently easy that the model consistently solves them correctly, so excluding them from additional updates is reasonable. (ii) Some instances are consistently incorrect, either because they exceed the current model’s capability or because potential label noise exists in the DAPO-Math-17K conversion process. In these cases, zero-advantage filtering can reduce the risk of reinforcing noisy or uninformative signals.

DAPO discards generated response groups with identical labels, so its response utilization is not directly comparable to GRPO and DCPO under the same RUR definition. We therefore evaluate DAPO’s training efficiency through the number of generated responses and the time required to complete the same number of update steps. To compensate for low sampling efficiency, DAPO uses a generation batch size dynamically adjusted between 1 and 4 times the training batch size. Across the four models, DAPO requires 3 to 5 times more generated responses than DCPO for the same number of parameter updates, at least doubling GPU hours without improving final performance in the reported comparisons. On Qwen2.5-14B, the proportion of DAPO responses used falls below 30% after about 300 steps, illustrating the generation overhead introduced by dynamic sampling.

Taken together, the *RUR* and efficiency analyses show that DCPO preserves substantially more response-level learning signal than GRPO while avoiding the regeneration cost incurred by DAPO. This improved use of generated data provides a direct empirical explanation for the performance gains reported in Table 1.

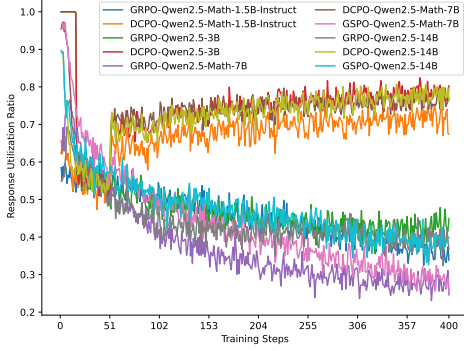


Figure 2: *RUR* progresses during training.

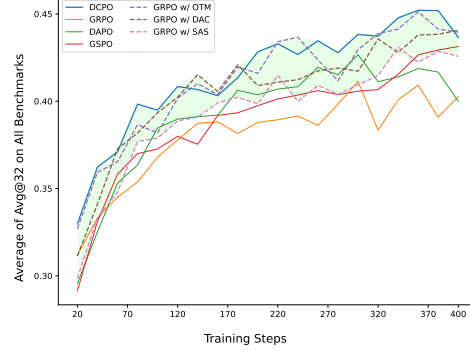


Figure 3: Component ablation on Qwen2.5-Math-7B using average Avg@32.

5.4 ABLATION STUDIES

We conduct an ablation study on Qwen2.5-Math-7B to isolate the contribution of each DCPO component, using **Avg@32** as the evaluation metric because it reflects the sampled policy distribution. Each ablation modifies one component of the GRPO framework while keeping the remaining settings fixed; the KL divergence term is removed to align the objective family with DAPO, GSPO, and full DCPO. Figure 3 reports learning curves from 20 to 400 steps.

The ablation curves show that OTM, SAS, and DAC each improve the GRPO baseline, while their combination yields the strongest and most stable trajectory. Baseline performance tends to plateau in later training, whereas full DCPO continues to improve. The study is limited to Qwen2.5-Math-7B and an aggregate Avg@32 metric, but within this setting it indicates that the three components contribute complementary benefits rather than a single component accounting for the entire gain.

(i) **Only-Token-Mean loss (OTM)**: Averaging the loss within each response rather than across the whole batch improves over GRPO by preserving response-level advantage relationships. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing stepwise advantage normalization with cumulative smoothed advantage improves GRPO and approaches DAPO, supporting the role of cross-step reward statistics in stabilizing optimization. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping boundary with the probability-adaptive boundary yields a clear improvement and produces a more stable curve than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining the three mechanisms achieves the best overall ablation result, consistent with the design premise that token-level clipping, response-level advantage utilization, and loss aggregation address different inefficiencies.

Overall, the ablation study supports the claim that DCPO’s performance gain arises from the coordinated effect of its three components. Together with the *TCR* and *RUR* analyses, these results link the method’s mechanism to improved data efficiency and stability in mathematical RLVR training.

6 CONCLUSION

In this work, we attempt to address key limitations of existing policy optimization methods: restricted token-level exploration due to fixed cropping in low-probability regions, and significantly low sample efficiency due to the current step’s reward standardization. To this end, we introduced **Dynamic Clipping Policy Optimization (DCPO)**, a novel reinforcement learning pipeline that enhances the reasoning capabilities of large language models through two key innovations: (i) a **dynamically adaptive token-level clipping** mechanism that adjusts bounds based on old policy probabilities, enabling more efficient exploration of rare tokens with low probabilities, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates the cumulative reward distribution across optimization steps, mitigating reward randomness from high-entropy sampling. Extensive

experiments on four mathematical reasoning benchmarks with four models demonstrate DCPO’s effectiveness. For example, the DCPO-7B variant achieved an Avg@32 score of 38.8 on the challenging AIME24 benchmark, outperforming GRPO (32.1) and DAPO (31.6). In addition, DCPO increased the response utilization ratio by 28% over GRPO, doubled the training efficiency compared to DAPO, and reduced the token clipping ratio by an order of magnitude. These results highlight DCPO’s ability to significantly improve data utilization and diversity exploration in RLVR. Future work will explore the extension of DCPO to other domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In previous works, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for the token t in the response j is calculated by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards of the G responses generated in the i -th step, as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for the same prompt are identical, responses with zero advantages do not contribute to model update, resulting in response waste.

A.3.1 GRPO

Shao et al. (2024) proposed the **Group Relative Policy Optimization (GRPO)**, which first samples G responses for each query, assigns rewards R through a rule-based reward function and estimates token-level advantages as in Equation (11). Finally, the model parameters are updated using Equation (12), referred to as **Sequence Level Mean loss (SLM)** as:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty constrains the policy from deviating too far from a reference policy (e.g., a pre-trained model). The fixed clipping threshold ε prevents overly large updates by limiting the change in token probability ratios. This averaging strategy, though widely adopted, inherently dilutes per-response importance.

A.3.2 DAPO

Similarly, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, which observed that in GRPO, longer sequences exert a disproportionate influence on gradient update. They standardize token counts across the entire batch, leading to the **Token Level Mean loss (TLM)** as Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. It discards the prompts whose all responses share identical reward, and regenerates responses to maintain batch size, leading to severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can inject high-variance noise into gradient estimates. This noise tends to accumulate over long sequences and is further amplified by the clipping operation itself. GSPO alleviates the problem by replacing token-wise clipping with sequence-wise clipping, which yields a more stable and effective training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)} \right) \right)$$

GSPO discards more than 10% of responses with non-zero advantages, resulting in high data waste and training efficiency. Although GRPO discards responses with sequence-level high variance, it still retains a large number of responses with token-level high entropy, resulting in training instability.

A.4 REWARD CALCULATION

To provide a verifiable and interpretable supervision signal, we adopt a rule-based reward model that evaluates each generated response based on the correctness of the answers and the compliance with the format, as Equation (15). Specifically, a reward of +1 is assigned when both the format and the answer are correct; a reward of 0 is given when the format is correct but the answer is incorrect; and a reward of -1 is assigned when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j is the index of the response group for a given problem in the i -th training step, and the reward is calculated based on the format and correctness of the response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning to stabilize policy optimization and prevent large, destabilizing parameter update. This mechanism is particularly important when using importance sampling to estimate expected values under a new policy from samples generated by an old policy.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ by importance sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be significantly inflated, which is a common challenge in importance sampling. The true variance of the function $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference between the importance-sampled variance and the true variance highlights the potential for inflation, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To mitigate this difference, many prior works, including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025), employ a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

However, a fixed clipping bound inherently constrains the policy’s capacity for exploration, particularly problematic in regions where the old policy assigns a low probability mass. In such cases, the absolute bound for low probabilities is severely limited, restricting the potential for meaningful policy updates. While methods like DAPO attempt to address this by introducing asymmetric clipping bounds, the fundamental limitation of a non-adaptive constraint remains.

In addition, recent studies such as Yue et al. (2025) suggest that the reasoning paths leveraged by RLVR models are largely present within the sampling distribution of the base model. This is further evidenced by the minimal parameter shift observed during off-policy batch update, as reflected in the small fraction of clipped parameters. Given this stability, we propose a more practical alternative to constrain the probability ratio $r(x)$ through the dynamic-adaptive mechanism by including the probability in the restriction as in Equation (21), replacing the conventional fixed clipping approach of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Expanding the substitution of $p(x) = r(x)q(x)$ into the inequality of a previous equation yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, we must adhere to the fundamental constraints of probability distributions and their ratios. Specifically, as shown in Equation (23), the probabilities $p(x)$ and $q(x)$ must be non-negative and less than or equal to 1. Consequently, the probability ratio $r(x)$ must also be non-negative, and the clipping hyper-parameter ϵ must be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure the validity and practical applicability of the clipping bound derived from Equation (22), we define the necessary conditions for its bounds as outlined in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

As we can get from Equation (22) to Equation (24), the probability ratio $r(x)$ should be met by the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered around the ratio value of 1 to minimize the discrepancy of bounds between the two conditions and to maintain consistency with the conventional fixed bound of GRPO, and define the following conditions to ensure the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, to ensure that the square number is not less than zero, we use the method of comparing with 0 and taking the maximum value of the approximation like $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. Finally, we can get the following conditions, as **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds in Inequality 27, we set different upper and lower clipping thresholds ϵ_{high} and ϵ_{low} hyperparameters to align the clipping bound with the GRPO at specific points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To ensure that our dynamic adaptive clipping bounds align with the fixed symmetric clipping bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we derive the specific values for ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

Typically, in previous practices(e.g., GRPO), the fixed clipping threshold ϵ is set to 0.2, and we can get our clipping hyperparameters to be approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$, while $\epsilon_{low} \neq \epsilon_{high}$ has a different purpose from the asymmetric fixed clipping setting in DAPO.

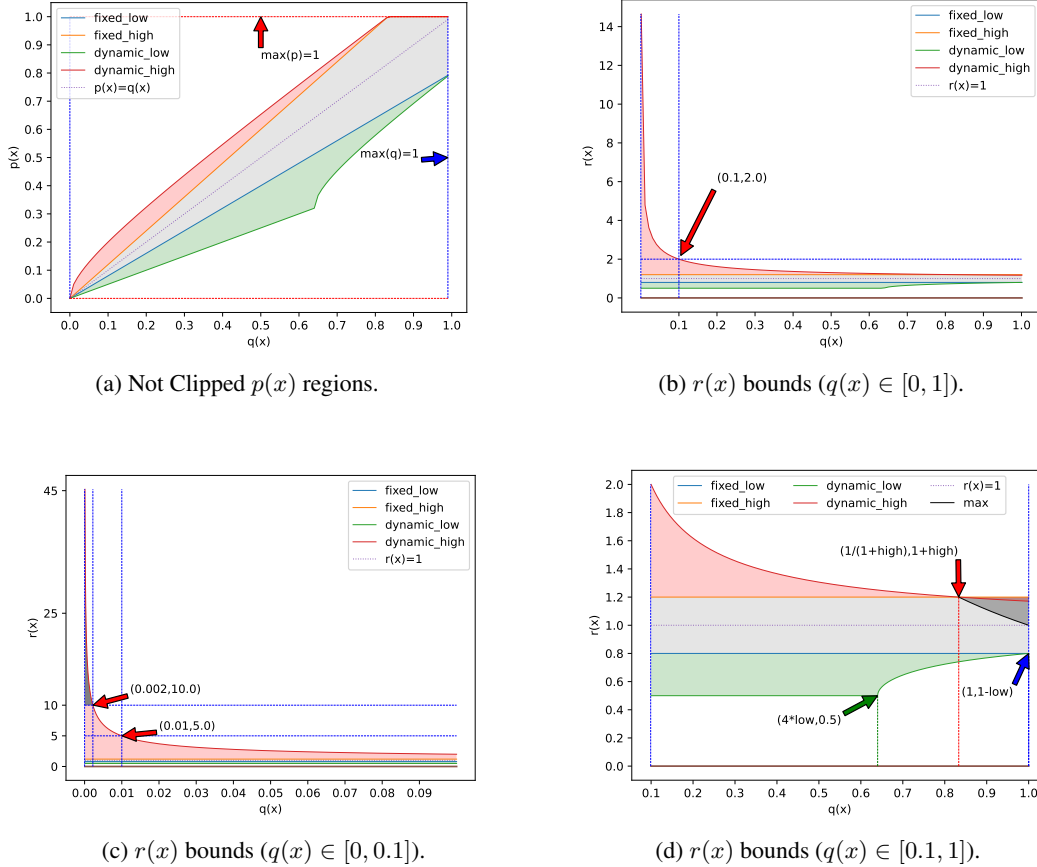


Figure 4: Clipping bound comparisons. Lines show bounds for fixed clipping ($\epsilon = 0.2$) vs. dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$).

Dynamic versus fixed clipping: Figure 4 illustrates how the bound behaves under the fixed clipping scheme used in GRPO ($[1 - \epsilon, 1 + \epsilon]$) and under our probability-aware adaptive clipping. For a token with an old probability $q(x)$, the fixed bound provides a constant relative clipping width 2ϵ . Consequently, as $q(x)$ decreases, the absolute exploration space allowed shrinks linearly (Figure 4a). In contrast, the relative width of dynamic adaptation is inversely proportional to $q(x)$, so the exploration space remains roughly constant throughout the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the admissible interval $[1 - \varepsilon, 1 + \varepsilon]$ of the fixed clipping scheme is independent of $q(x)$, so the model’s ability to explore low-probability tokens does not improve, while the dynamic adaptive bound expands rapidly (Figure 4b), resulting in a larger relative space for rare tokens, which is exactly the behavior required for efficient RL.

Behavior for extremely low probabilities: Figure 4c shows the clipping interval when $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the upper bound for $r(x)$ of dynamic-adaptive clipping increases rapidly inversely proportional to the square root, while the corresponding value for fixed clipping remains constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we enforce a hard ceiling $r_{\max} = 10$ to prevent the resulting instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the upper limit of adaptive clipping is set to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$ the dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas the fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$. When $q(x)$ increases beyond this threshold, the dynamic-adaptive lower bound smoothly contracts towards the fixed value. Similarly, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$, and merges with it for larger $q(x)$. Thus, the adaptive interval $[dynamic_{low}(q(x), \varepsilon_{low}), dynamic_{high}(q(x), \varepsilon_{high})]$ provides a probability-dependent widening of the admissible region for rare tokens while preserving the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) contains both a positive and a negative advantage term. During the operation with minimum loss value, the ratio $r(x)$ for a positive advantage is clipped in the interval $[0, dynamic_{high}(q(x), \varepsilon_{high})]$. For a negative advantage, the clipping interval $[dynamic_{low}(q(x), \varepsilon_{low}), \infty)$. Due to an excessively large $r(x)$, which can destabilize training, we limit the ratio for both positive and negative advantages at $r_{\max} = 10$. This precaution is inspired by Ye et al. (2020) and ensures that the adaptive limits do not produce excessively high importance weights.

In short, regardless of positive or negative advantages, compared to fixed clipping, dynamic adaptive clipping provides the model with more space to explore diversity when rare tokens have a low probability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration where prompts are used to generate responses and update the model parameters. We define the following statistics for reward normalization.

For the average for the rewards of responses, we define the equations as the group of functions as in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G * (i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For the variance for the rewards of the responses, we define the equations as the group of functions as in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage, for the calculation of the value of σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations of μ_{new}^i , σ_{new}^i , μ_{old}^i and σ_{old}^i instead of direct calculations of original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

We use the following template for all experiments in this paper, which is based on the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We set a symmetric clipping threshold of $\epsilon = 0.2$ in GRPO and lower/upper clipping thresholds (0.2, 0.28) with an additional 512-token soft-punish cache and the maximum probability ratio 10 for negative advantage for DAPO, lower/upper clipping thresholds ($3e - 4, 4e - 4$) for GSPO, and low/high clipping thresholds (0.16, 0.2) for DCPO, preserving the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d), and set the maximum probability ratio to 10 for both positive and negative advantages, different from DAPO.

Training was conducted for 400 steps with a batch size of 512 for response generation and a mini-batch size of 32 for parameter update. We used a temperature equal to 1 and top_p equal to 1 to generate $G=16$ responses for each problem, while the prompt max length was set to 1,024 and the response max length was set to 3,072, which is much smaller than the maximum length of 20k used in the original DAPO.

We adapted our training codebase from Verl (Sheng et al., 2024) on 32xH20 GPUs. Unless stated otherwise, the hyperparameters matched those of DAPO. For loss computation, we retained the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted average formulation from GRPO or DAPO, instead of the approximate weighted calculation used in Verl, to ensure compatibility with different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 shows the performance of the model using the average Avg@1 metric. This metric is calculated by averaging the greedy decoding (Avg@1) results on the MATH500, AMC23, AIME24, and AIME25 benchmarks and represents the best performance of the model. As can be seen in the figure, DCPO consistently outperformed GRPO and DAPO across all four models,

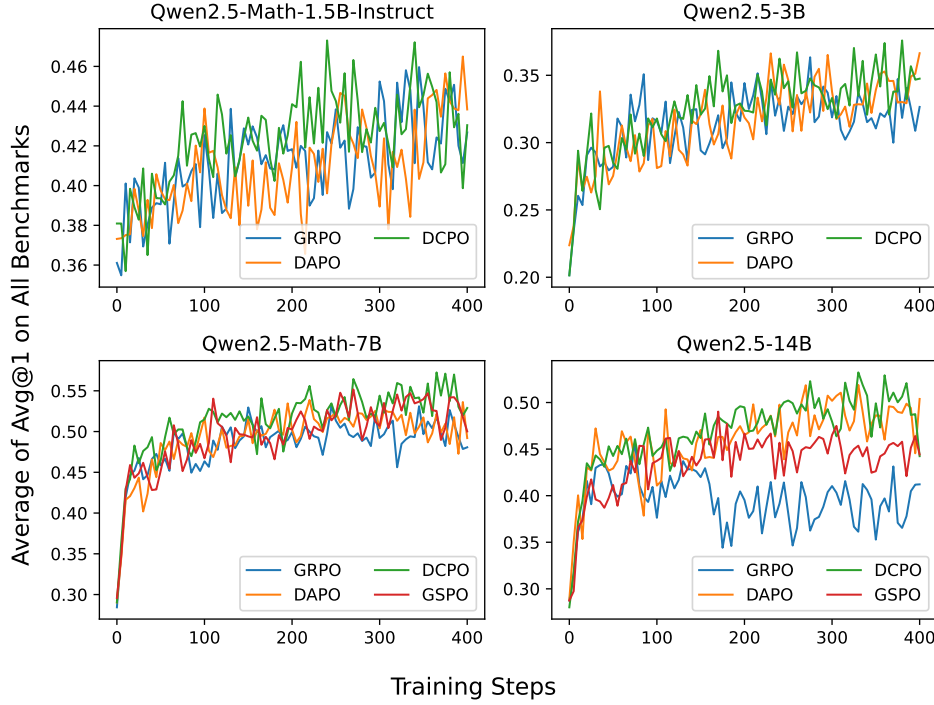


Figure 5: Avg@1 performance across benchmarks

and outperformed GSPO on the two models where GSPO was evaluated. Furthermore, its overall performance was more stable than the baseline, without the significant performance drop seen in GRPO or the significant fluctuations seen in DAPO. This suggests that DCPO can stimulate stronger model reasoning capabilities during training, providing greater potential for the model to explore even better performance.

Avg@32 Trend Figure 6 shows the Avg@32 performance of DCPO with GRPO, DAPO and (where available) GSPO at different model sizes on AMC23, AIME24, and AIME25 benchmarks, clearly demonstrating DCPO’s exceptional robustness and stability. In contrast, GRPO exhibits significant fluctuation and occasional performance collapse. By GRPO, the Qwen2.5-14B model even experiences a significant performance drop after approximately 170 training steps. Across all evaluated models (from 1.5B to 14B), the DCPO performance trajectory exhibits a consistently upward trend with minimal fluctuation. Even in the early stages of training, DCPO maintains the fastest growth rate and maintains its upward trend. Furthermore, compared to DAPO, DCPO achieves superior performance while saving half of training time and resources. These experimental results demonstrate the effectiveness and robustness of dynamic-adaptive clipping and smoothing advantage standardization to improve the exploration at the response level and the label level of diversity.

Entropy Trend Figure 7 shows that GRPO exhibits a sharp entropy decay after an initial warm-up phase and quickly stabilizes in a flat, low-entropy range. This entropy collapse reflects a rapid loss of policy randomness, reducing the diversity of sampled responses, and thus limiting subsequent policy optimization.

The behavior of DAPO and DCPO is model-dependent. In the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. In contrast, for the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the trends differ significantly. DAPO maintains the highest entropy value, showing a clear upward trend accompanied by large fluctuations. On Qwen2.5-Math-7B, the entropy value of GSPO is similar to that of DCPO, and the average metric performance is also comparable. However, on Qwen2.5-14B, the performance of GSPO exhibits a fluctuating downward trend, resulting in poor performance. In contrast, DCPO reaches a moderate

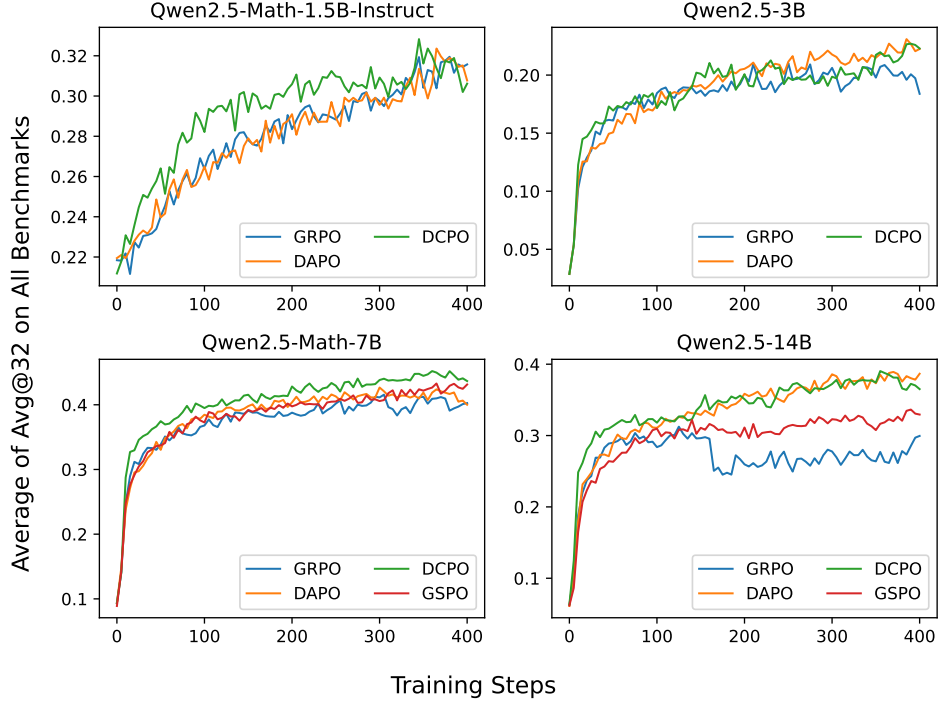


Figure 6: Avg@32 performance across benchmarks.

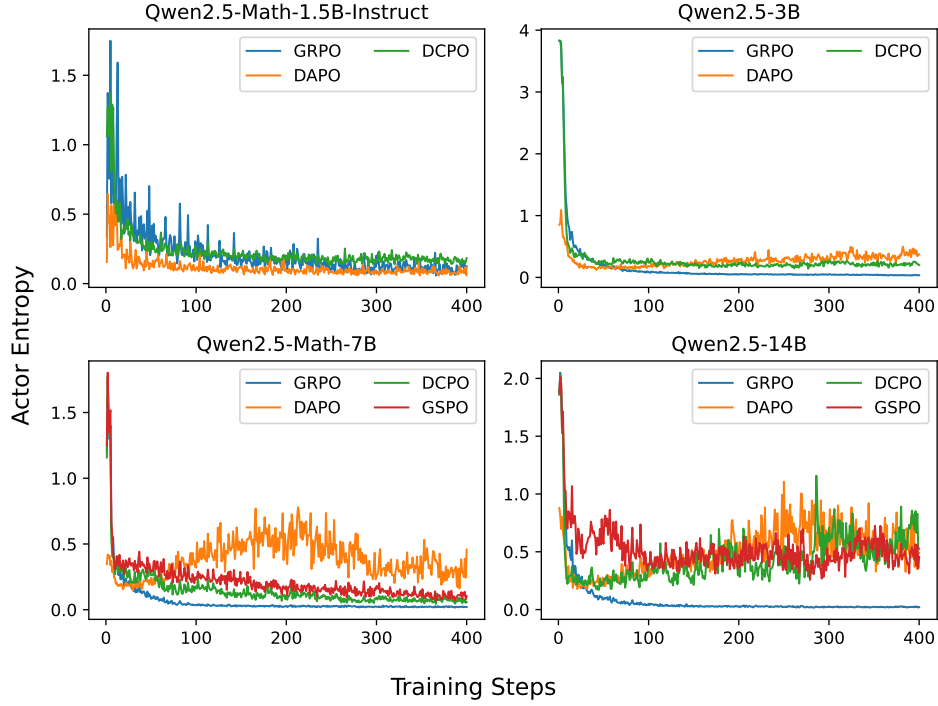


Figure 7: The Entropy of the models during Training.

entropy level: its curve is significantly smoother and remains confined to a bounded band between the low-entropy region of GRPO and the high-variance region of DAPO.

These observations suggest a U-shaped relationship between entropy and training performance. Excessively low entropy deprives the learner of exploratory signal, leading to premature convergence, while excessively high or unstable entropy compromises training stability. DCPO maintains entropy within a moderate and well-balanced range, promoting convergence and sufficient exploration, a benefit that is particularly pronounced as model capacity grows. Therefore, DCPO’s entropy regulation mechanism may underlie its superior inference performance and enable more robust policy learning across models of varying sizes.